A7 — Individual Submission Report

Student Name: Sawitta Thiabsaeng
Student ID: 6731503123
Project / Capability Name: Support Request Classifier
A6 Project Reference: Support Request Classifier — A6: Build and Integrate an AI Decision Capability
Implementation Method: Jev Playground / Other (local Python fallback and evaluation scripts)

1. Grounding Context

Knowledge Source:

The project uses a local knowledge base stored in `knowledge_base.json`. It contains six support-related knowledge records covering account issues, password recovery, website and server errors, application problems, support hours, and general services.

Knowledge Records:

ID	Knowledge Topic	Purpose
KB01	Account and login issues	Identify account-related problems
KB02	Password recovery	Support password recovery requests
KB03	Website and server errors	Identify technical problems
KB04	Application crashes and bugs	Identify application problems
KB05	Support opening hours	Answer questions about support hours

KB06	General information and services	Handle general inquiries

Retrieval / Selection Method:

The `retrieve_context.py` script uses keyword matching to select relevant records based on words in the user's support message. This method is simple to implement and understand, but it may miss vague messages or alternative phrases that do not match the expected keywords.

Context Supplied to Model:

Example of grounding instructions:

> You are a support request classifier. Use the knowledge base to interpret the user's message. KB01 covers account and login issues; KB02 covers password recovery; KB03 covers website and server errors; KB04 covers application crashes and bugs; KB05 covers support hours; and KB06 covers general services. Classify the message into ACCOUNT_ISSUE, TECHNICAL_ISSUE, or GENERAL_INQUIRY. If the message is ambiguous or lacks information, do not make an unsupported assumption.

Manual Grounding Experiment:

Test message: “My password reset page is not loading.”

Measurement	**Without Context**	**With Context**
Selected category	TECHNICAL_ISSUE	TECHNICAL_ISSUE
ACCOUNT_ISSUE probability	27%	36%
TECHNICAL_ISSUE probability	73%	64%
GENERAL_INQUIRY probability	0%	0%

Displayed latency	1,071 ms	548 ms

The selected category remained the same, while the displayed probabilities changed. Only one message was tested per condition, so this experiment does not establish improved classification accuracy or lower latency.

Evidence: `A7_grounding_without_context.png` and `A7_grounding_with_context.png`.

2. Evaluation Dataset

The dataset is stored in `eval-cases.json` and contains 10 evaluation cases. The table uses T01–T10 to match the template. The corresponding original JSON identifiers are E01–E10.

ID	Case Type	Input	Expected Output	Expected Review/Fallback	Reason
T01	Normal	I forgot my password and cannot log in.	ACCOUNT_ISSUE	ROUTE_TO_SUPPORT	Clear password and login issue
T02	Normal	The website shows a 500 Internal Server Error.	TECHNICAL_ISSUE	ROUTE_TO_SUPPORT	Clear website error
T03	Normal	What are your support	GENERAL_INQUIRY	ROUTE_TO_SUPPORT	General support question

		opening hours?			
T04	Edge	I cannot log in.	ACCOUNT_ISSUE	ROUTE_TO_SUPPORT	Short but identifiable login issue
T05	Edge	The app crashes when I open it.	TECHNICAL_ISSUE	ROUTE_TO_SUPPORT	Application crash
T06	Edge	Can you tell me what services are available?	GENERAL_INQUIRY	ROUTE_TO_SUPPORT	General service inquiry
T07	Fallback	It does not work. Please help.	N/A	ASK_FOR_MORE_INFORMATION	Insufficient details
T08	Fallback	Empty input	N/A	ASK_FOR_MORE_INFORMATION	No message content

T09	Difficult	I cannot access my account, and the website keeps showing an error.	N/A	SEND_TO_HUMAN	Multiple issues and ambiguity
T10	Difficult	My password reset page is not loading.	N/A	SEND_TO_HUMAN	Password recovery with technical failure

Note: N/A means that a single category is not the primary expected output for that case; the expected fallback action is the evaluation target. Verify the exact expected values against `eval-cases.json` before submission.

3. Baseline Evaluation

Measurement Method:

The original fallback implementation was evaluated using `python .\evaluate.py`. The benchmark used 10 cases with 5 repetitions per case, resulting in 50 runs.

Individual per-case latency values were not available in the recorded summary, so they are marked N/A rather than estimated.

ID	Expected Output	Actual Output	Pass/Fail	Latency (ms)
T01	ACCOUNT_ISSUE / ROUTE_TO_SUPPORT	ACCOUNT_ISSUE / ROUTE_TO_SUPPORT	PASS	N/A

T02	TECHNICAL_ISSUE / ROUTE_TO_SUPPORT	TECHNICAL_ISSUE / ROUTE_TO_SUPPORT	PASS	N/A
T03	GENERAL_INQUIRY / ROUTE_TO_SUPPORT	GENERAL_INQUIRY / ROUTE_TO_SUPPORT	PASS	N/A
T04	ACCOUNT_ISSUE / ROUTE_TO_SUPPORT	ACCOUNT_ISSUE / ROUTE_TO_SUPPORT	PASS	N/A
T05	TECHNICAL_ISSUE / ROUTE_TO_SUPPORT	TECHNICAL_ISSUE / ROUTE_TO_SUPPORT	PASS	N/A
T06	GENERAL_INQUIRY / ROUTE_TO_SUPPORT	GENERAL_INQUIRY / ROUTE_TO_SUPPORT	PASS	N/A
T07	ASK_FOR_MORE_INFORMATION	ASK_FOR_MORE_INFORMATION	PASS	N/A
T08	ASK_FOR_MORE_INFORMATION	ASK_FOR_MORE_INFORMATION	PASS	N/A
T09	SEND_TO_HUMAN	SEND_TO_HUMAN	PASS	N/A
T10	SEND_TO_HUMAN	SEND_TO_HUMAN	PASS	N/A

Baseline Accuracy: 10 / 10 = 100% local action agreement

p95 Latency: 0.063 ms (approximate local benchmark value)

Cost per Task: N/A — model API and token costs were not measured.

Measurement Method:

The baseline evaluation used the original implementation and the 10-case evaluation dataset. The benchmark executed 50 runs (10 cases × 5 repetitions). The recorded baseline average local latency was 0.043 ms, with an approximate p95 latency of 0.063 ms.

The 100% result represents agreement between local action outputs and expected actions in this dataset. It does not represent actual AI model classification accuracy. The benchmark excludes model inference and network latency.

4. One Controlled Improvement

Selected Improvement:

Expand the deterministic fallback rule to recognize three additional descriptions of a broken password-reset or password-recovery page:

- `is broken`
- `does not work`
- `doesn't work`

Reason for Choosing It:

Before the change, the input “The password reset page is broken.” returned `ROUTE_TO_SUPPORT` instead of the intended `SEND_TO_HUMAN` action.

Before Configuration:

`fallback_before.py` preserves the original implementation. Its page-failure phrase list did not include the three additional phrases.

After Configuration:

`fallback.py` and `fallback_after.py` contain the additional phrases in the page-failure matching logic. The change allows the fallback system to recognize more ways of describing a failed password-recovery page and route the case for human review.

5. After-Improvement Evaluation

The original 10-case dataset was evaluated again after the improvement. The recorded local action agreement remained 10/10 cases. Individual per-case latency measurements were not available in the summary.

ID	Expected Output	Actual Output	Pass/Fail	Latency (ms)
T01	ACCOUNT_ISSUE / ROUTE_TO_SUPPORT	ACCOUNT_ISSUE / ROUTE_TO_SUPPORT	PASS	N/A
T02	TECHNICAL_ISSUE / ROUTE_TO_SUPPORT	TECHNICAL_ISSUE / ROUTE_TO_SUPPORT	PASS	N/A
T03	GENERAL_INQUIRY / ROUTE_TO_SUPPORT	GENERAL_INQUIRY / ROUTE_TO_SUPPORT	PASS	N/A
T04	ACCOUNT_ISSUE / ROUTE_TO_SUPPORT	ACCOUNT_ISSUE / ROUTE_TO_SUPPORT	PASS	N/A
T05	TECHNICAL_ISSUE / ROUTE_TO_SUPPORT	TECHNICAL_ISSUE / ROUTE_TO_SUPPORT	PASS	N/A
T06	GENERAL_INQUIRY / ROUTE_TO_SUPPORT	GENERAL_INQUIRY / ROUTE_TO_SUPPORT	PASS	N/A
T07	ASK_FOR_MORE_INFORMATION	ASK_FOR_MORE_INFORMATION	PASS	N/A
T08	ASK_FOR_MORE_INFORMATION	ASK_FOR_MORE_INFORMATION	PASS	N/A
T09	SEND_TO_HUMAN	SEND_TO_HUMAN	PASS	N/A

T10	SEND_TO_HUMAN	SEND_TO_HUMAN	PASS	N/A

Additional Targeted Tests

These are additional tests for the new phrases and are separate from the original 10-case dataset.

Test ID	Input	Expected Action	Result
T09	The password reset page is broken.	SEND_TO_HUMAN	PASS
T10	The password reset page does not work.	SEND_TO_HUMAN	PASS
T11	The password recovery page is broken.	SEND_TO_HUMAN	PASS

All three additional targeted tests passed in the recorded results. They are not included in the original 10-case agreement score.

6. Before / After Comparison

Metric	Before	After	Change
Accuracy / local action agreement	100% (10/10)	100% (10/10)	No change
p95 Latency	0.063 ms	0.079 ms	+0.016 ms
Cost per Task	N/A	N/A	Not measured

Cases That Improved:

The targeted input “The password reset page is broken.” returned `ROUTE_TO_SUPPORT` before the change and `SEND_TO_HUMAN` after the change. The two additional targeted inputs also passed after the improvement.

Cases That Became Worse:

No reduction was observed in the original 10-case local action agreement. It remained 100%. However, the dataset is small and cannot establish that no regressions occur for other inputs.

Trade-offs:

The change improves handling of the tested alternative phrases but does not increase the original 10-case agreement score. Average local latency increased from 0.043 ms to 0.050 ms, while approximate p95 latency increased from 0.063 ms to 0.079 ms. These small timing differences may reflect measurement noise and do not represent end-to-end AI latency. Model API cost was not measured.

Conclusion:

The improvement is worthwhile as a targeted coverage and safety fix because it recognizes additional descriptions of a failed password-recovery page. The available evidence does not show increased accuracy on the original dataset or improved AI model accuracy.

7. Engineering Reflection

What I learned:

I learned that an AI capability should be evaluated using a repeatable test dataset rather than only a few successful examples. Deterministic fallback rules can request more information or route ambiguous cases to a human instead of making unsupported decisions. Targeted tests are also useful for checking alternative phrasings of the same problem.

Why a few successful examples are insufficient:

Real support messages may be incomplete, ambiguous, or contain multiple issues. A small test dataset may miss alternative phrasings and cannot establish general AI model accuracy.

What I would evaluate next:

I would expand the dataset with more normal, ambiguous, and difficult support messages. I would also evaluate retrieved context across multiple inputs, measure classification metrics using human-labelled ground truth, test invalid or conflicting model outputs, gather reviewer feedback, and measure end-to-end latency and API costs if a model API is integrated.

8. Evidence and References

Evidence / Source	Purpose
`knowledge_base.json`	Grounding knowledge records
`retrieve_context.py` and `a7_retrieval.txt`	Retrieval implementation and output
`eval-cases.json`	Original 10-case evaluation dataset
`a7_before_evaluation.txt`	Baseline evaluation evidence
`a7_after_evaluation_final.txt`	After-improvement evaluation evidence
`a7_before_benchmark.txt`	Baseline benchmark evidence
`a7_after_benchmark_final.txt`	After-improvement benchmark evidence
`a7_after_targeted_tests.txt`	Additional targeted test results
`a7_after_fallback_tests.txt`	Fallback test results

`A7_grounding_without_context.png`	Manual experiment without context
`A7_grounding_with_context.png`	Manual experiment with context
`fallback_before.py`	Original fallback implementation
`fallback_after.py` and `fallback.py`	Updated fallback implementation

AI Assistance Disclosure:

AI assistance was used to help organize and format this report. The measurements and outcomes should be checked against the submitted execution logs and screenshots.

9. Submission Checklist

- ☑ Same capability as Week 08 — Support Request Classifier
- ☑ At least 5 knowledge records — KB01–KB06
- ☑ 10 evaluation cases with expected results — `eval-cases.json`
- ☑ Baseline results completed — Section 3
- ☑ Exactly one improvement applied — expanded password-recovery page failure phrase matching
- ☑ After-improvement results completed — Section 5
- ☑ Before/after comparison included — Section 6
- ☑ Actual evidence and reflection listed — Sections 7–8